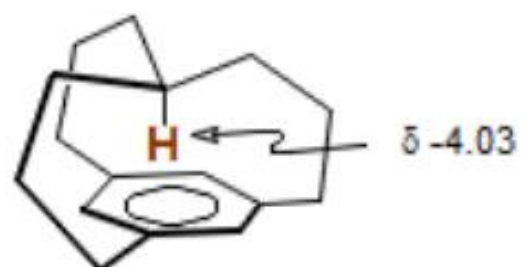
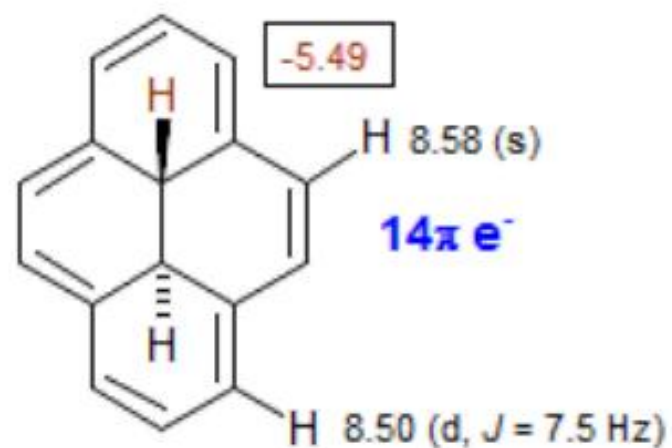


The alkene region and the benzene region

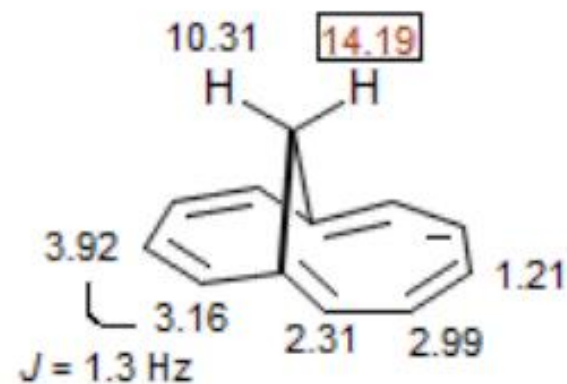
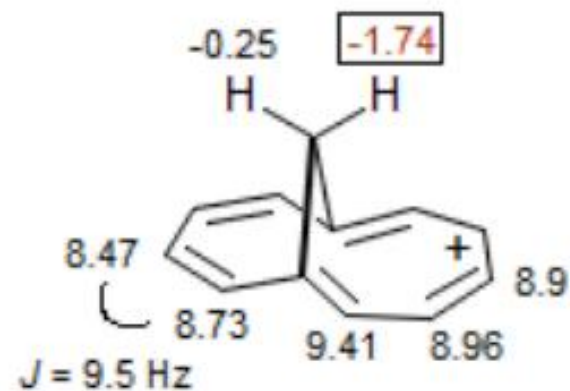
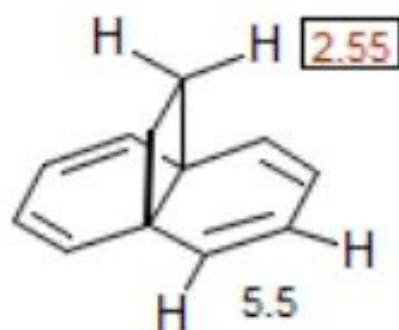
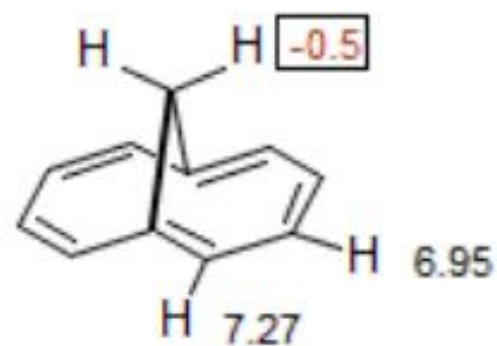
In ^{13}C NMR, alkene and benzene carbons came in the same region of the spectrum, but in the ^1H NMR spectrum the H atoms attached to arene C and alkene C atoms sort themselves into two groups. Look at the ^{13}C and ^1H chemical shifts of cyclohexene and benzene. The two carbon signals are almost the same (1.3 ppm difference, < 1% of the total 200 ppm scale) but the proton signals are very different (1.6 ppm difference = 16% of the 10 ppm scale).



Pascall *JACS* **1987**, 6878



Boekelheide *JACS* **1970**, 3511



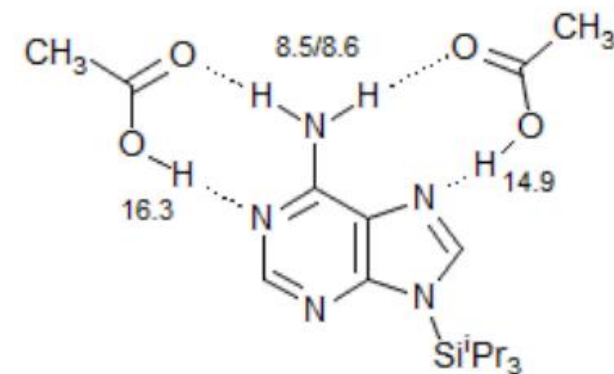
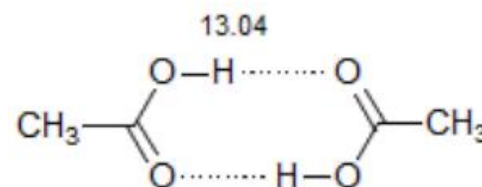
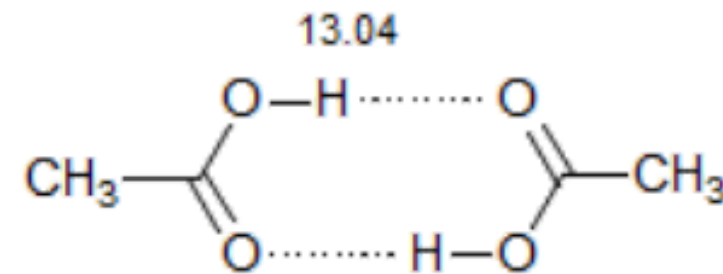
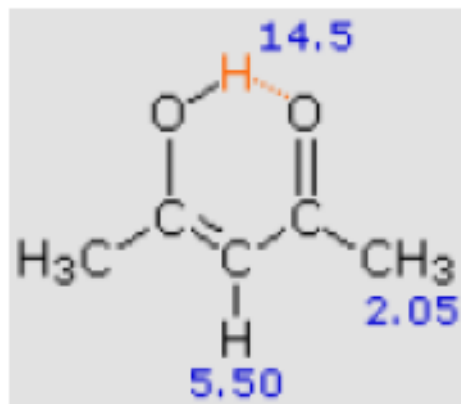
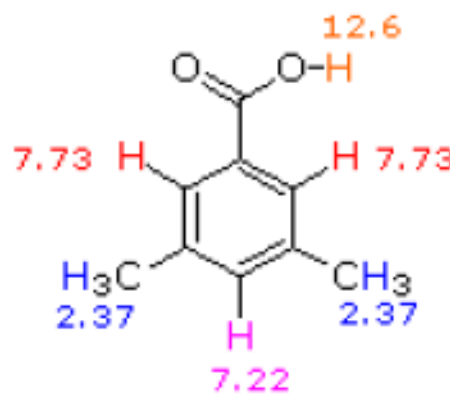
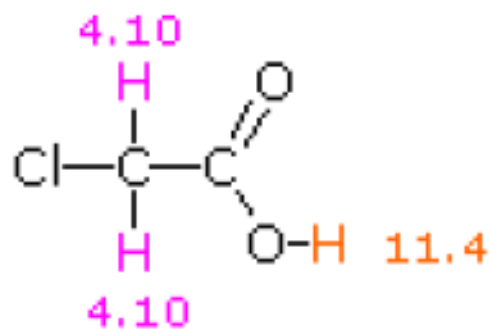
Hydrogen Bonding

In general hydrogen bonded protons show higher value of chemical shift than the non-hydrogen bonded protons. Due to the high electro-negativity of the atom to which proton is hydrogen bonded, the electron cloud around the proton is decreased and thus causes deshielding of the proton. The downfield shift of the absorption depends on the strength of the hydrogen bonding. Greater is the degree of hydrogen bonding of proton, greater the downfield shift of a proton.

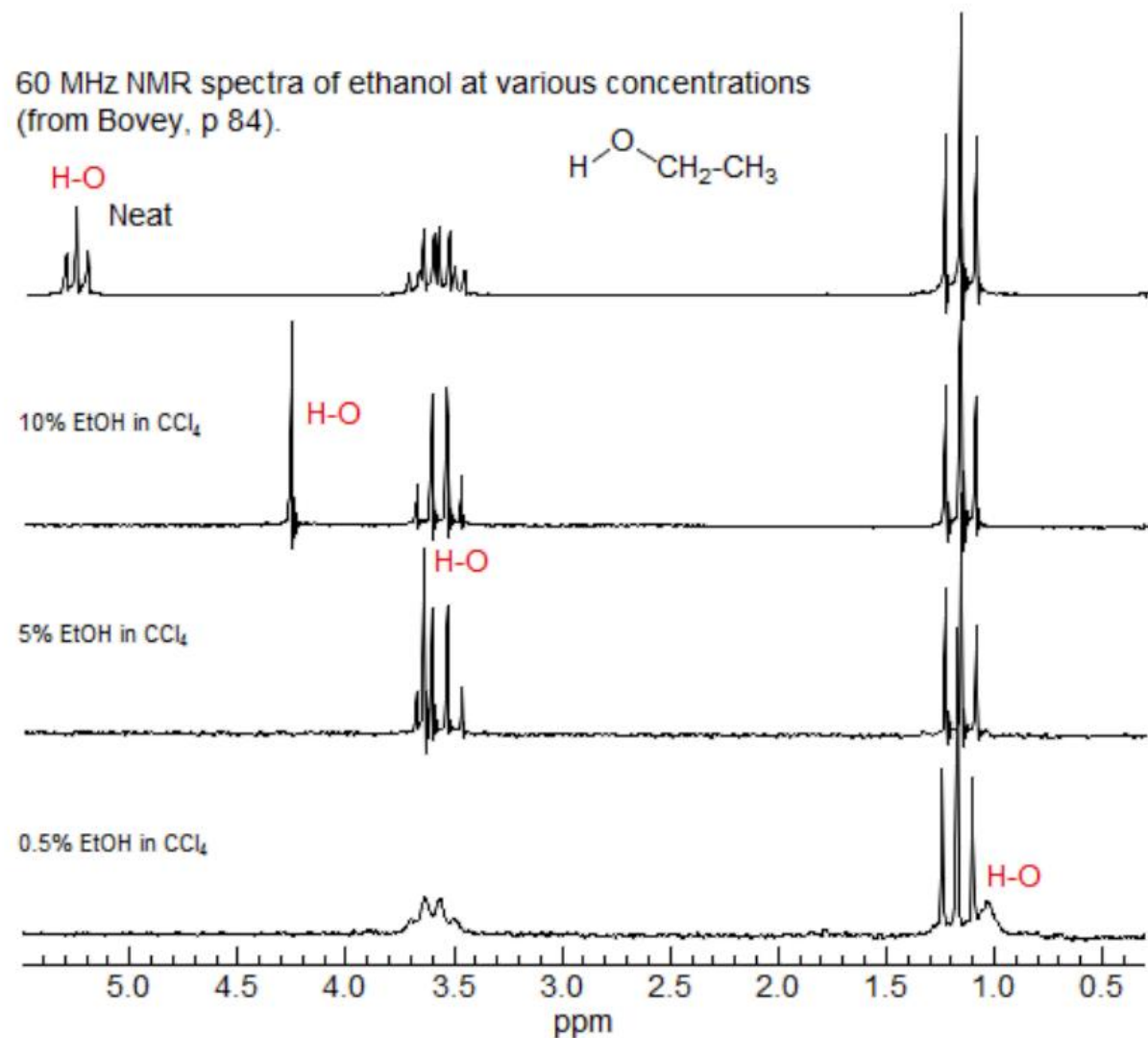
This is due to hydrogen bonding that the chemical shifts of OH, NH and SH protons vary over a wide range. It is unpredictable as the extent of hydrogen bonding depends on concentration, temperature, solvent and compound structure. The OH signals tend to move downfield at higher compound concentration because of increased hydrogen bonding. Also on increasing temperature, hydrogen bonding decreases and hence OH signals moves slightly upfield. Both OH and NH signals move downfield in H-bonding solvents like DMSO or acetone.

The intermolecular and intramolecular hydrogen bonding can be distinguished by the use of ^1H NMR spectroscopy. The intermolecular hydrogen bonding depends on the concentration of the sample and decreases on dilution with a non-polar solvent and also with increasing temperature. The pure ethanol show signals near 5 ppm but on dilution with non hydrogen-bonded solvents (CCl_4 , CDCl_3 , C_6D_6), the OH signal shift upfield due to the breaking of intermolecular hydrogen bonds. The more acidic hydroxyl group of phenol also shows similar concentration dependence to that of alcohols.

The intramolecular hydrogen bonding does not depend on the concentration of the sample and hence show no change in their absorption position on dilution. The salicylates, enols of β -diketones and carboxylic acids are some classic examples of intramolecular hydrogen bonding. The hydroxyl proton of enol form of 2,4-pentanedione (4-hydroxypent-3-ene-2-one) shows a peak significantly down field (10-16 ppm) compared to the OH signal in other function groups due to strong intramolecular hydrogen bonding. Similarly, the hydroxyl proton of carboxylic acid displays a very downfield peak at about 10-13 ppm which may be attributed to the favored hydrogen-bonded dimeric association.

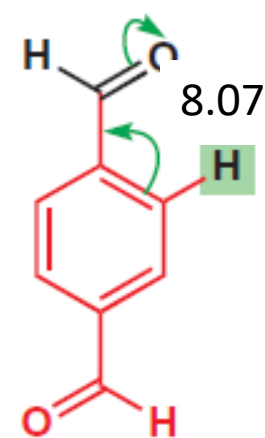
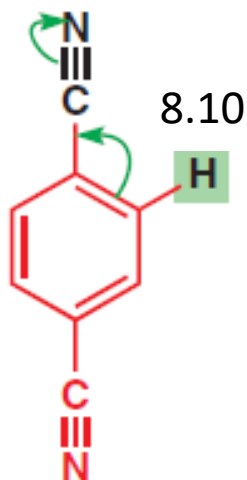
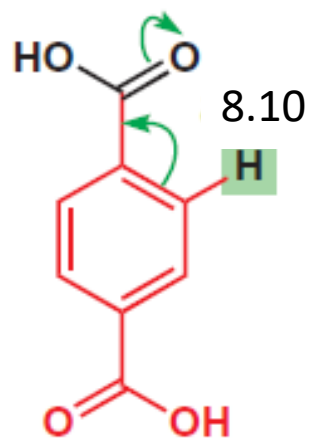
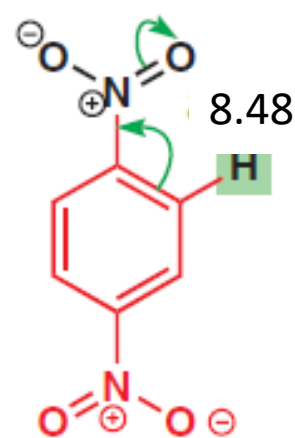


Alcohol OH Protons. In dilute solution of alcohols in non-hydrogen-bonding solvents (CCl_4 , CDCl_3 , C_6D_5) the OH signal generally appears at δ 1-2. At higher concentrations the signal moves downfield ($+\delta$) as a result of increased fraction of H-bonded alcohols, e.g. the OH signal of ethanol comes at δ 1.0 in a 0.5% solution in CCl_4 , and at δ 5.13 in the pure liquid (from Bovey).

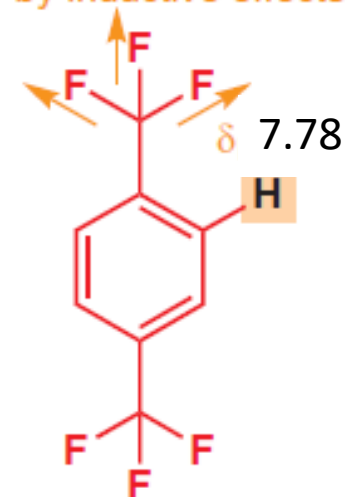


How electron donation and withdrawal change chemical shifts?

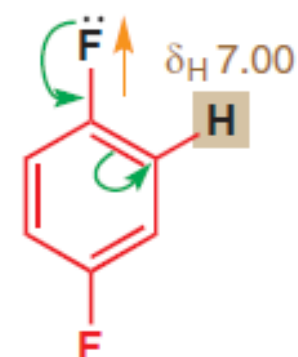
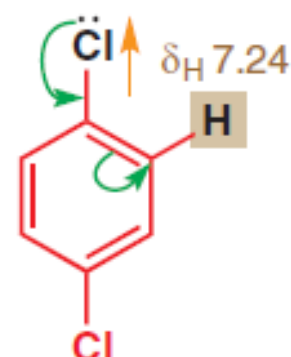
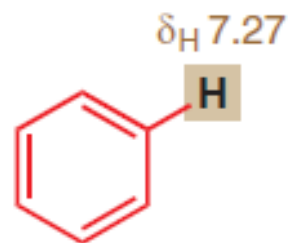
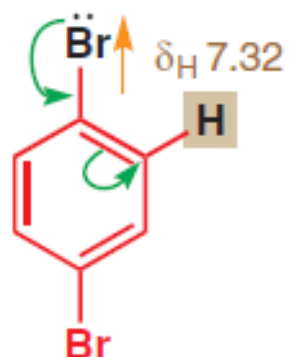
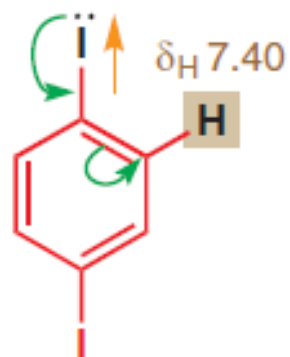
the effect of electron-withdrawing groups
by conjugation



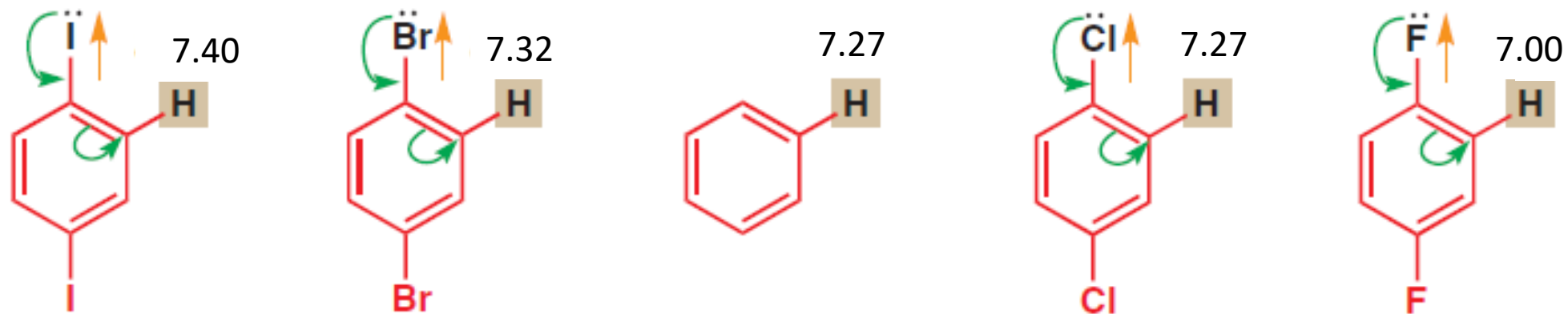
by inductive effects



balance between withdrawal by inductive effect and donation of lone pairs by conjugation

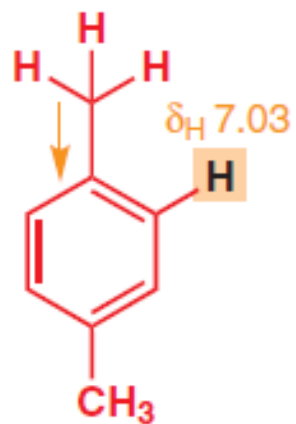


balance between withdrawal by inductive effect and donation of lone pairs by conjugation

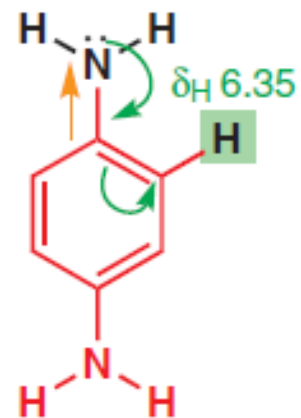
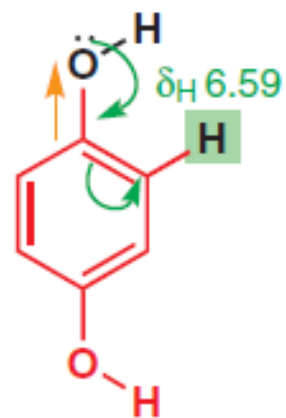
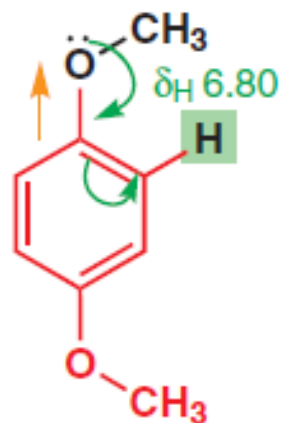


the effect of electron-donating groups

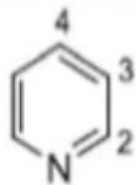
by inductive effect



balance between withdrawal by inductive effect and donation of lone pairs by conjugation—electron donation wins



Dec-2019 The correct match of ^{13}C NMR chemical shift values (δ ppm) for pyridine is



(1) C2: 136; C3: 124; C4: 150

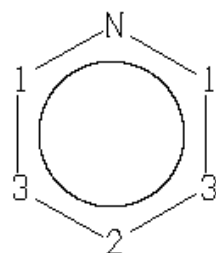
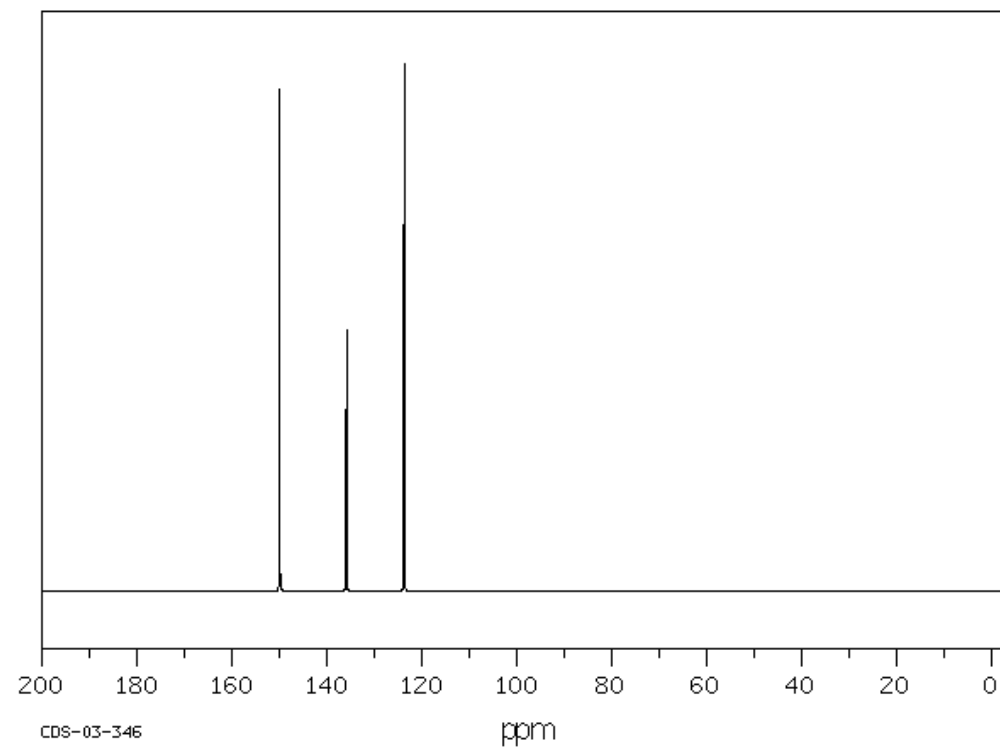
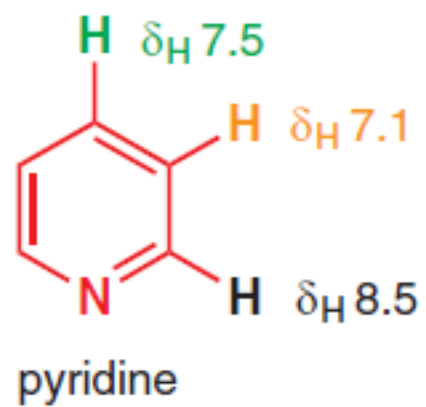
(2) C2: 124; C3: 150; C4: 136

(3) C2: 150; C3: 124; C4: 136

(4) C2: 150; C3: 136; C4: 124

25.16 MHz

18.4 wt% in CDCl₃

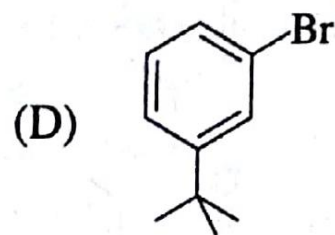
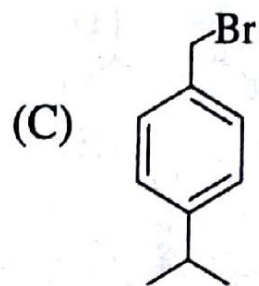
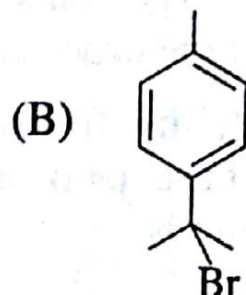
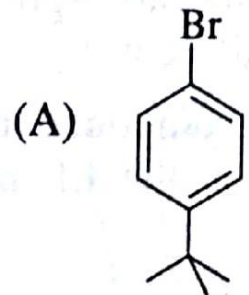


ppm	Int.	标记碳
149.94	950	1
135.89	493	2
123.75	1000	3

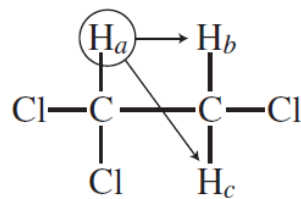
(NET_Dec, 2017)

46. The organic compound that displays following data is :

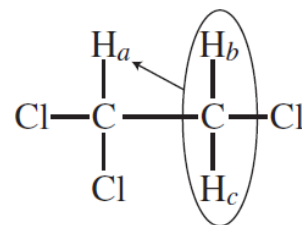
^1H NMR (400 MHz) : δ 7.38 (d), 7.25 (d), 1.29 (s) ppm



See

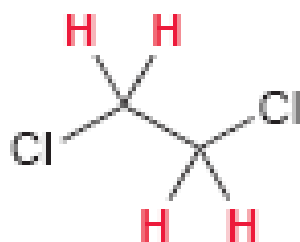


Two neighbors give a triplet
 $(n + 1 = 3)$ (area = 1)

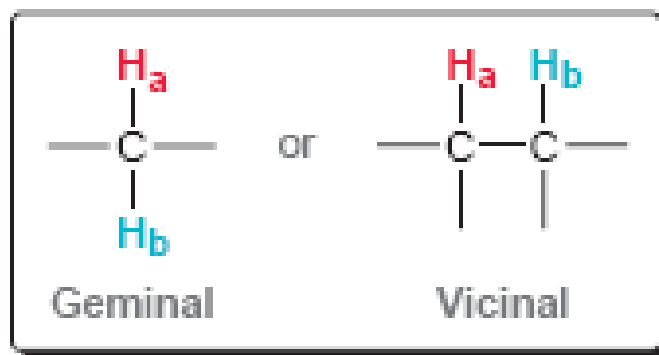


One neighbor gives a doublet
 $(n + 1 = 2)$ (area = 2)

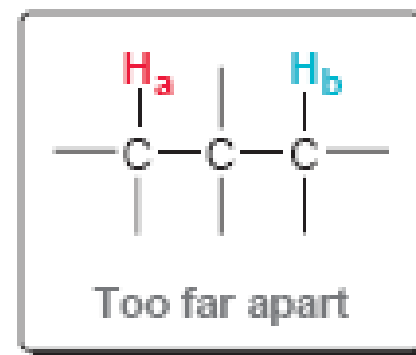
Equivalent protons
 behave as a group



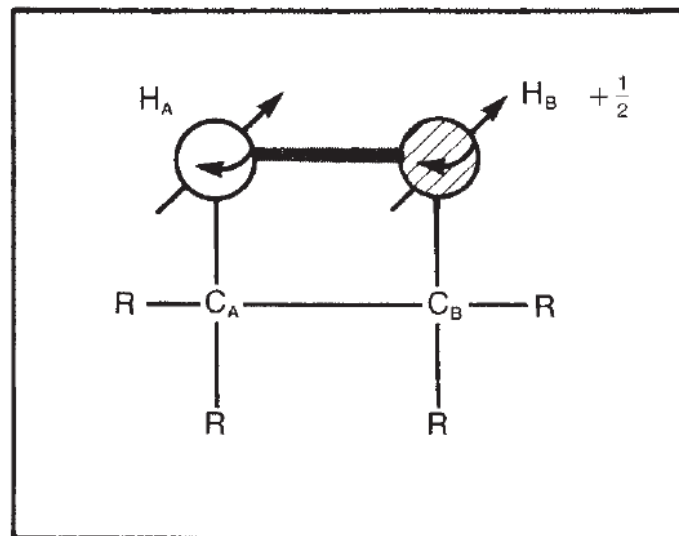
Four equivalent protons
 No splitting



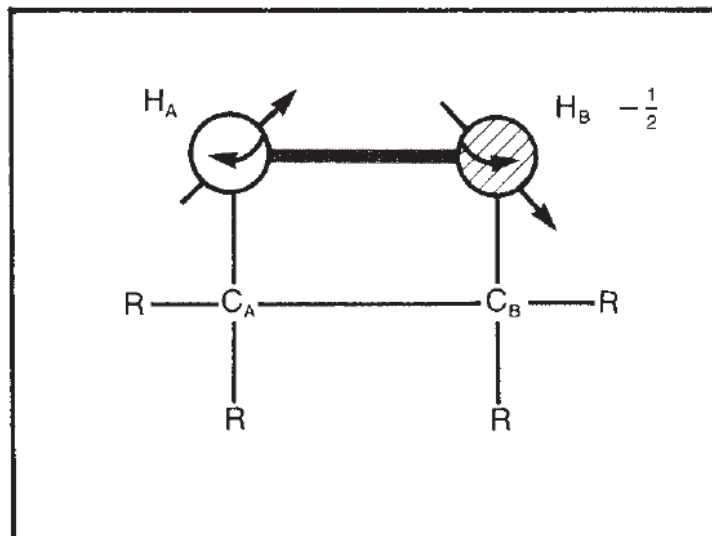
Splitting is observed



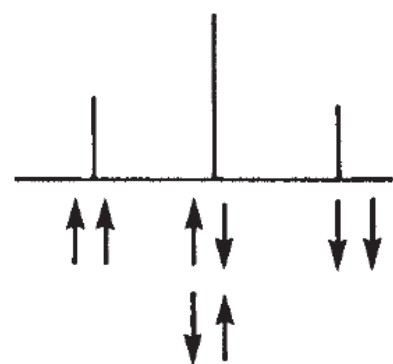
Splitting is generally not observed



X-type molecule

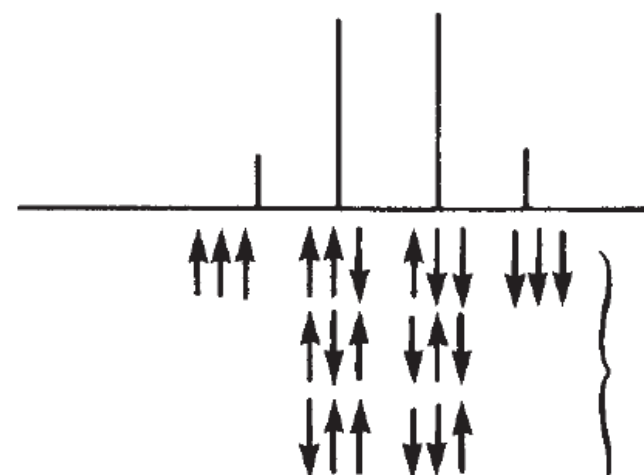


Y-type molecule



Possible spin arrangements
of the methyl protons

Net spin: +1 0 -1



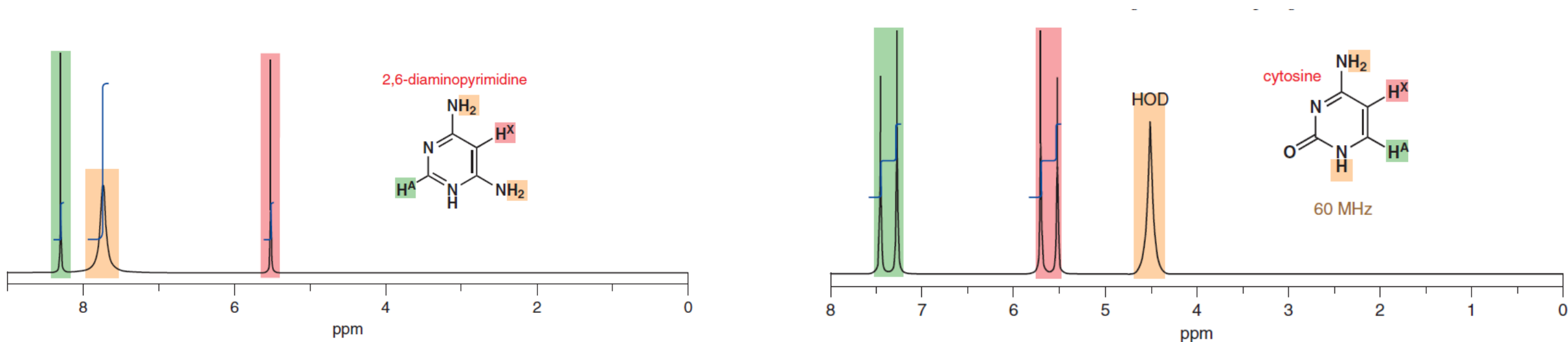
Possible spin arrangements
of the methyl protons

Net spin:

$+\frac{3}{2}$ $+\frac{1}{2}$ $-\frac{1}{2}$ $-\frac{3}{2}$

↑ = spin $+\frac{1}{2}$
↓ = spin $-\frac{1}{2}$

Spin-Spin Coupling



We have yet to discuss the real strength of proton NMR, something more important than chemical shifts and something that allows us to look not just at individual atoms but also at the way the C–H skeleton is joined together. This is the result of the interaction between nearby protons, known as *coupling*.

Cytosine vs 2,6-diaminopyrimidine:

Instead of giving just one line for each proton, they give two lines each—doublets. diaminopyrimidine: H^A is attached to an aldehyde-like C=N and so comes at lowest field. H^X is *ortho* to two electron-donating NH₂ groups and so comes at high field for an aromatic proton. These protons do not couple with each other because they are too far apart. They are separated by five bonds whereas the ring protons in cytosine are separated by just three bonds.

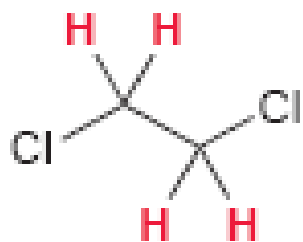
If the chemical shifts of the two signals differ significantly then the splitting patterns found in spectrum is easily recognizable. The patterns are symmetrically distributed on both sides of the proton chemical shift, and the central lines are always stronger than the outer lines. The most commonly observed patterns have been given descriptive names, such as doublet (two equal intensity signals), triplet (three signals with an intensity ratio of 1:2:1) and quartet (a set of four signals with intensities of 1:3:3:1) and so on. These patterns are given below.



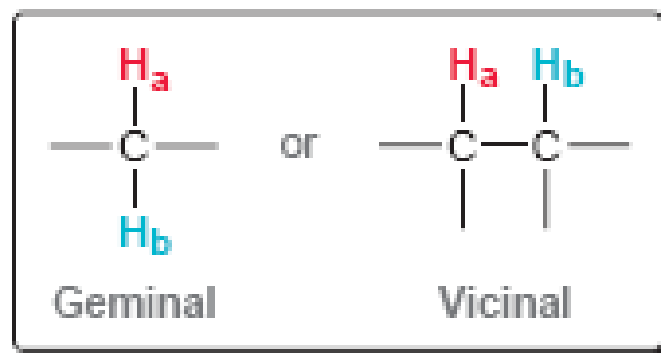
This spin-coupling is transmitted through the connecting bonds. For spin-coupling to be observed, the sets of interacting nuclei must be bonded in relatively close proximity (e.g. vicinal and geminal locations), or be oriented in certain optimal and rigid configurations. There are two major factors that determine whether or not splitting occurs:

1. Equivalent protons do not split each other. Consider the two methylene groups in 1,2- dichloroethane. All four protons are chemically equivalent, and therefore, they do not split each other. In order for splitting to occur, the neighbouring protons must be different than the protons producing the signal.

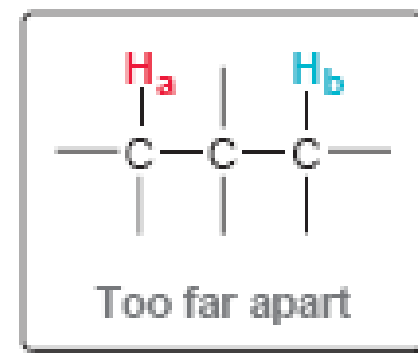
2. Splitting is observed for non-equivalent protons on the same carbon or adjacent carbons. When two protons are separated by more than three sigma bonds, splitting is generally not observed. Such long-range splitting is only observed in olefins, acetylenes, aromatics, heteroaromatics and in strained ring systems such as bicyclic compounds.



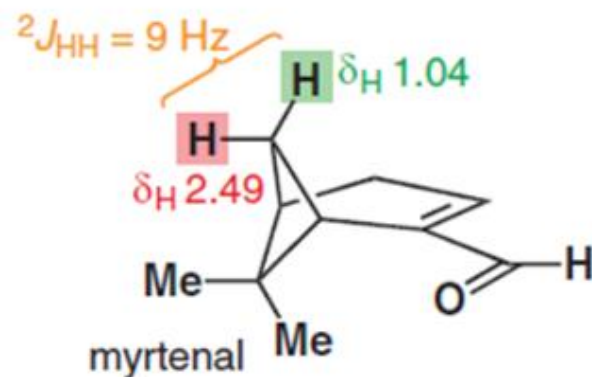
Four equivalent protons
No splitting



Splitting is observed

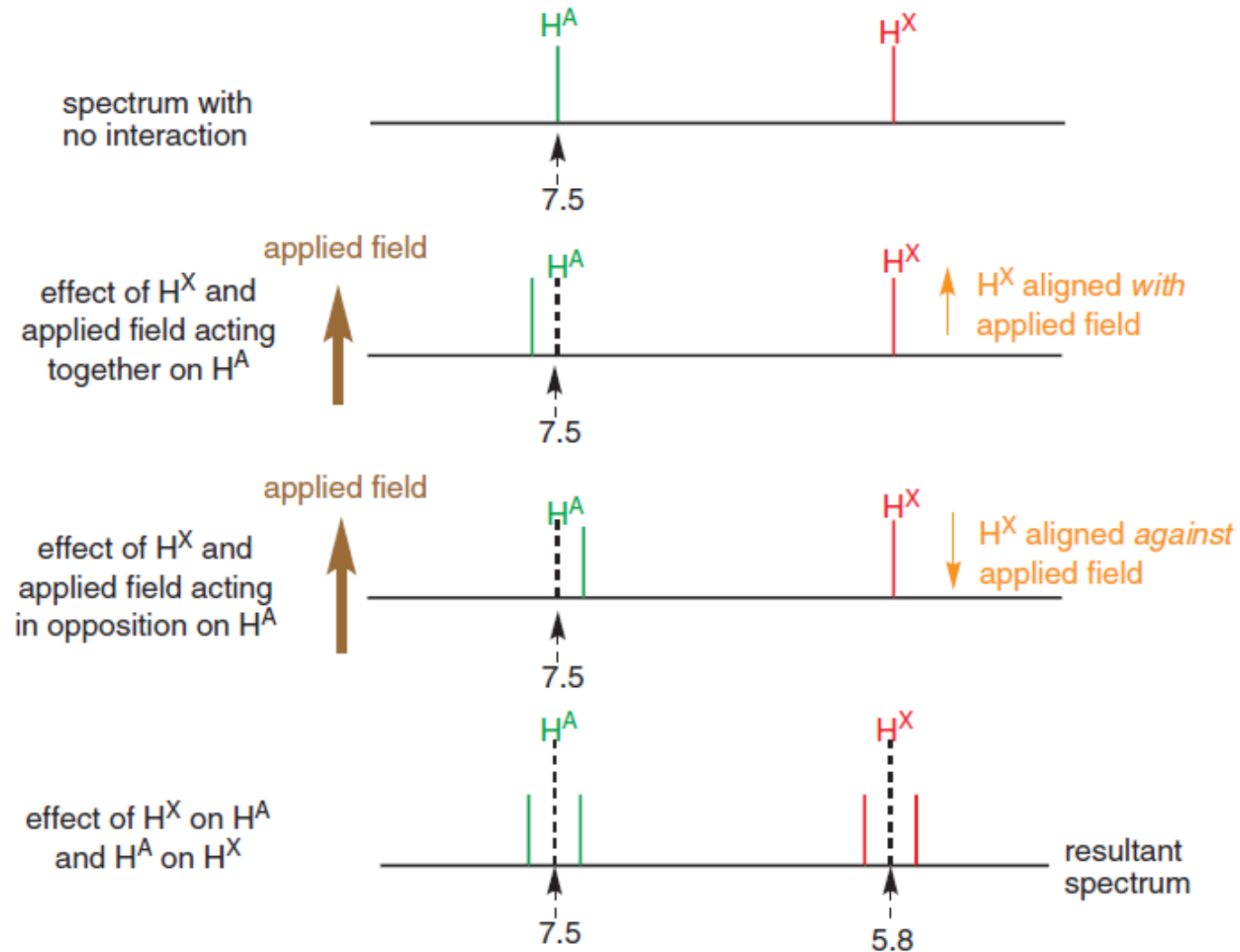


Splitting is generally not observed



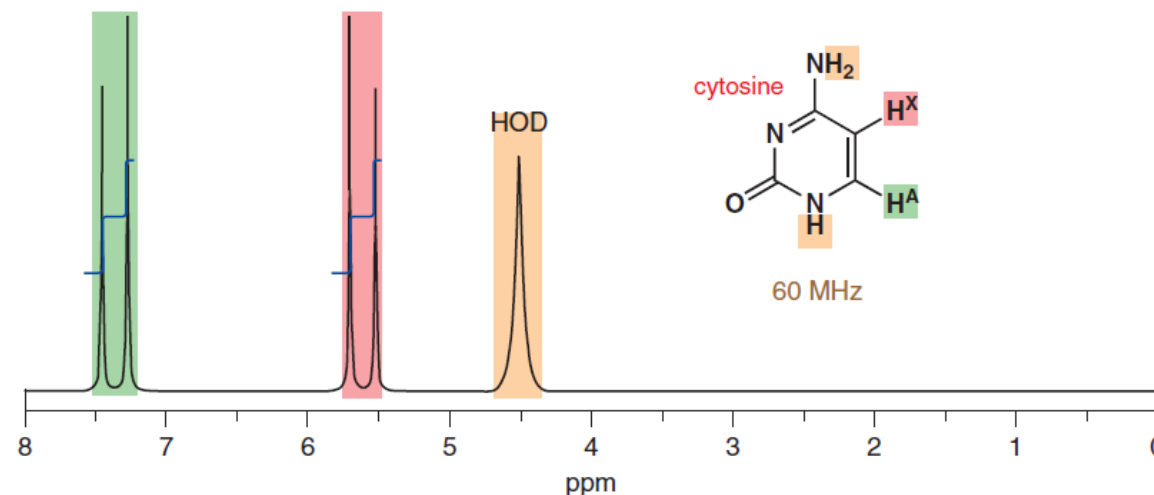
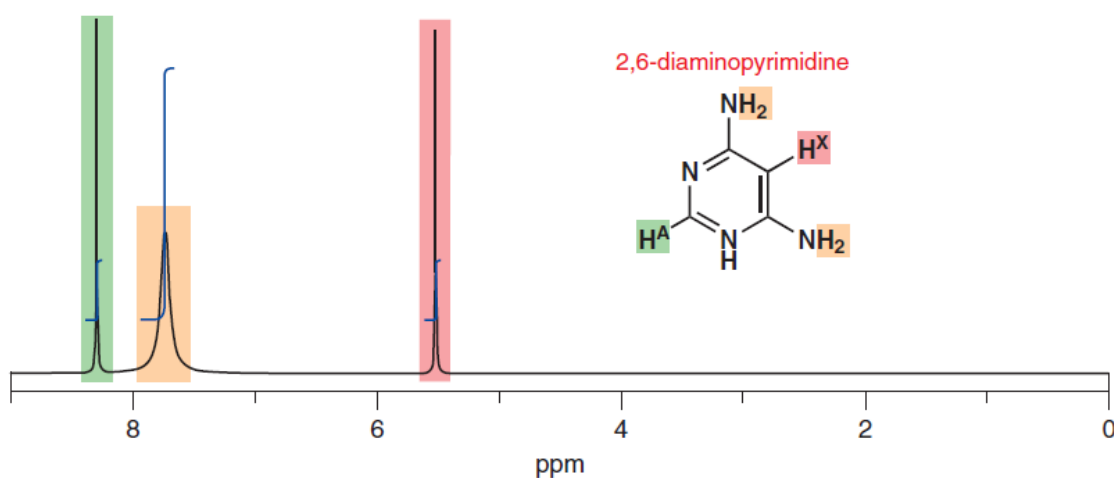
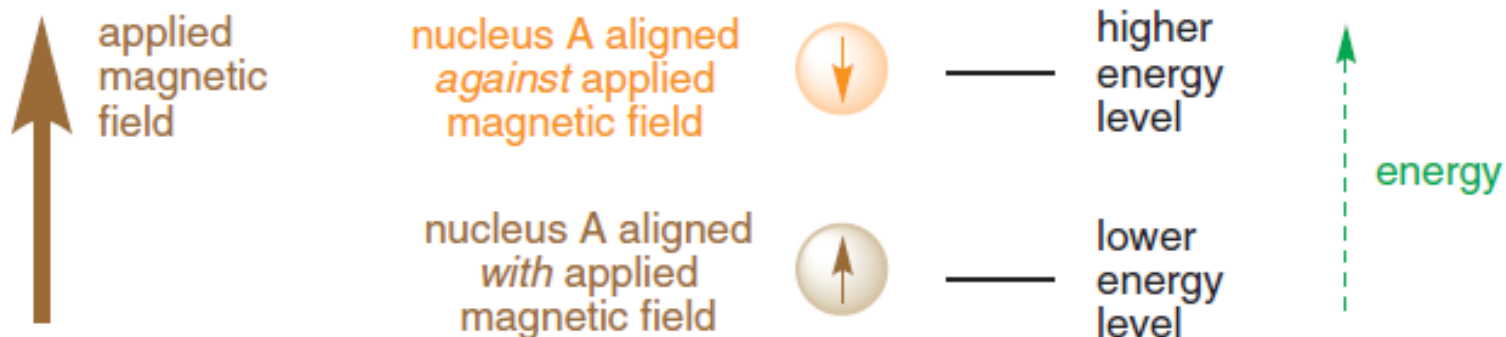
Origin of Splitting

First explanation: The diaminopyrimidine spectrum has two single lines (*singlets*) because each proton, H^A or H^X , can be aligned either with or against the applied magnetic field (only). The cytosine spectrum is different because each proton, say H^A , is near enough to experience the small magnetic field of the other proton H^X as well as the field of the magnet itself. The diagram shows the result.

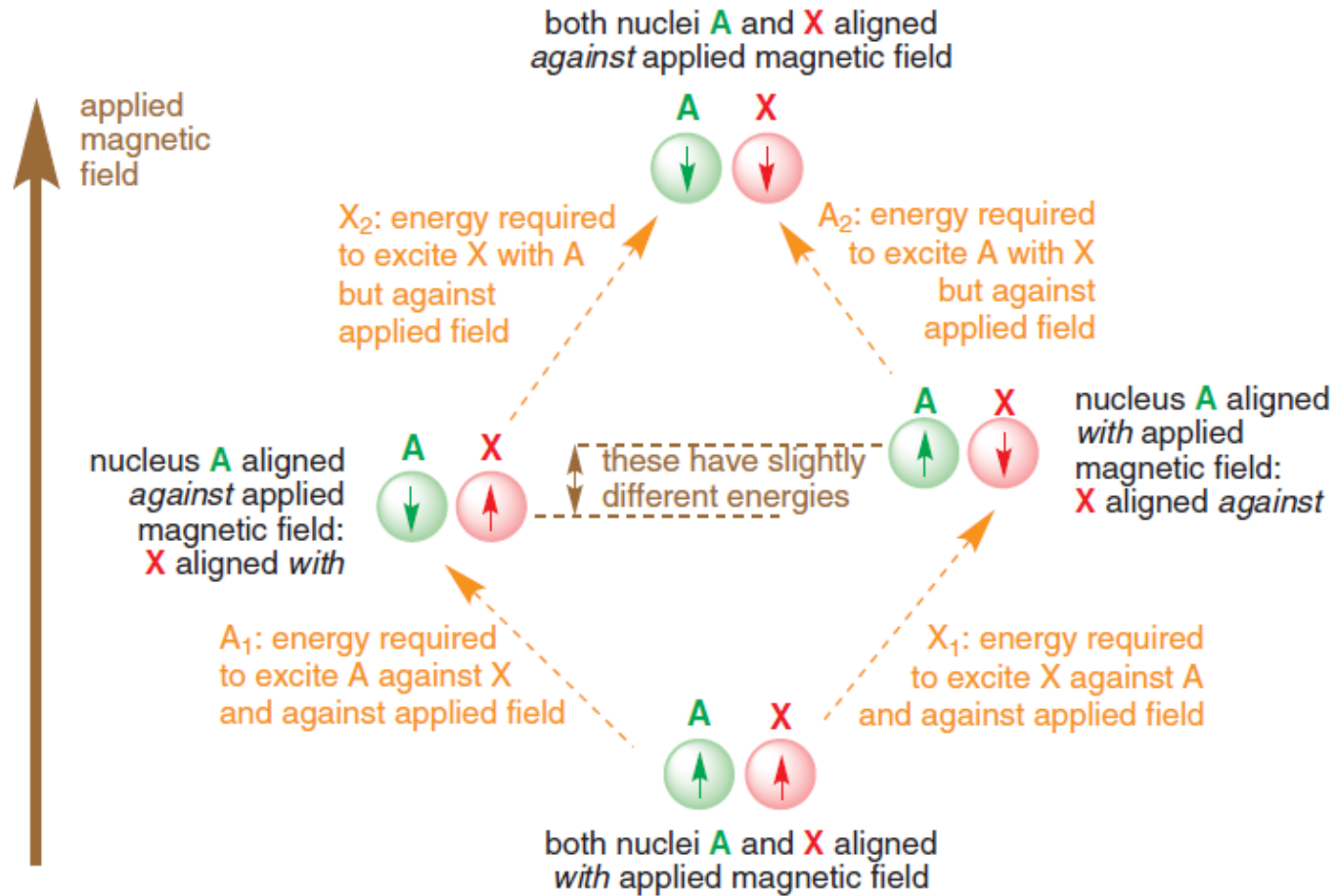


Second explanation: When hydrogen *nuclei* are near each other in a molecule, the nuclear energy levels also interact and split to produce new energy levels. If a single hydrogen nucleus interacts with a magnetic field: there are *two* energy levels as the nucleus can be aligned with or against the applied magnetic field, there is one energy jump possible, and there is a resonance at one frequency.

energy levels for one isolated nucleus

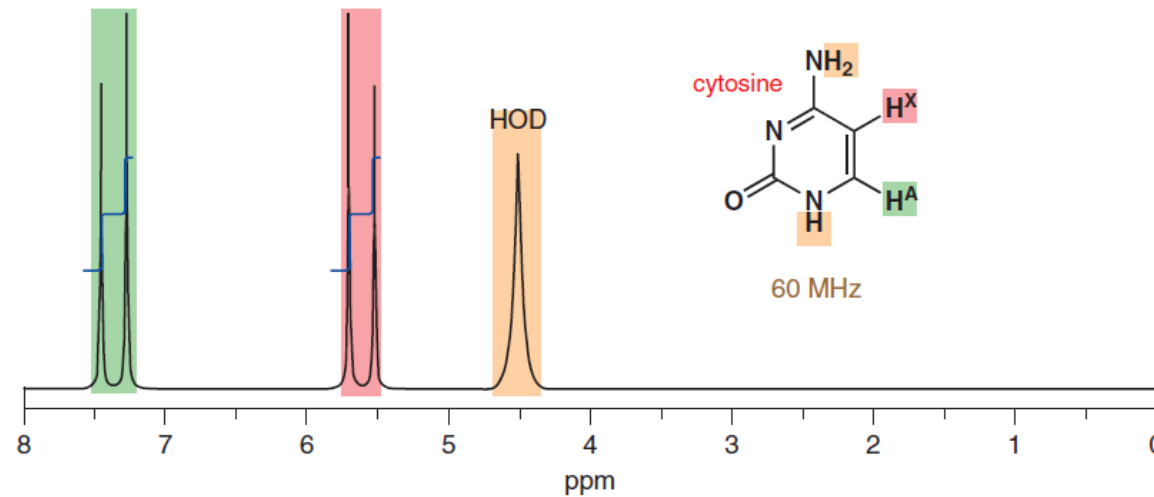


energy levels for two interacting nuclei



Cytosine: there are now *four* energy levels. Each nucleus H^A and H^X can be aligned with or against the applied field. There is one (lower) energy level where they are both aligned with the field and one (higher) level where they are both aligned against. In between there are two different energy levels in which one nucleus is aligned with the field and one against. Exciting H from alignment with to alignment against the applied field can be done in two slightly different ways, shown as A1 and A2 on the diagram. The result is two resonances very close together in the spectrum.

Please notice carefully that we cannot have this discussion about H^A without discussing H^X in the same way. If there are two slightly different energy jumps to excite H^A , there must also be two slightly different energy jumps to excite H^X . A_1 , A_2 , X_1 , and X_2 are all different, but the *difference* between A_1 and A_2 is exactly the same as the *difference* between X_1 and X_2 . Each proton now gives two lines (a doublet) in the NMR spectrum and the splitting of the two doublets is *exactly the same*. We describe this situation as coupling. We say 'A and X are coupled' or 'X is coupled to A' (and vice versa). We shall be using this language from now on.



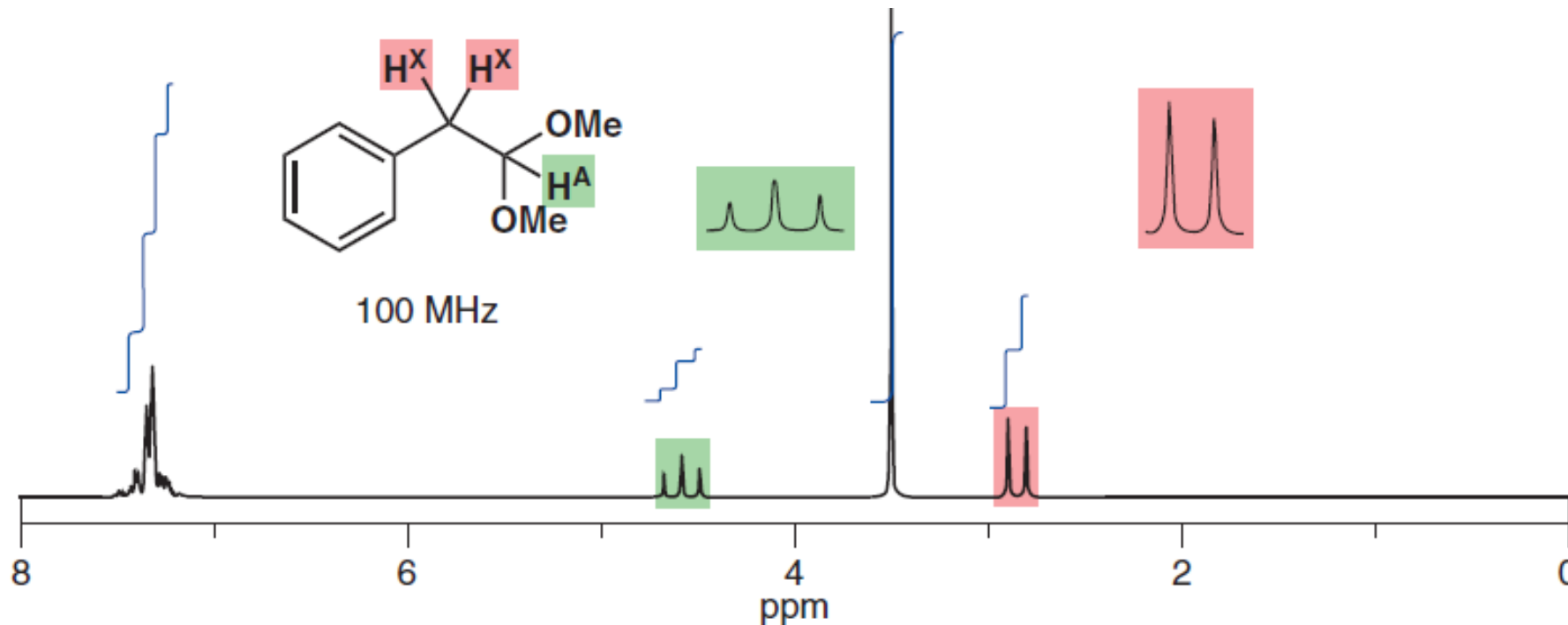
Cytosine: two doublets on the aromatic ring. Each is split by the same amount. The separation of the lines is the coupling constant and is called J . In this case $J = 4$ Hz. We measure chemical shifts in ppm because we get the same number regardless of the rating of the NMR machine in MHz. We measure J in Hz because we also get the same number regardless of the machine. The coupling constant unlike the chemical shift expressed in Hz, does not depend on the applied field strength. This is because the strength of the magnetic moment of a neighbouring proton, which is the source of the spin spin coupling phenomenon, does not depend on the applied field strength.

Measuring coupling constants in hertz

To measure a coupling constant it is essential to know the rating of the NMR machine in MHz (megahertz). This is why It is important to mention the machine rating along with the spectrum '400 MHz ^1H NMR spectrum'. Couplings may be marked on the spectrum, electronically, but if not then to measure the coupling, measure the distance between the lines by ruler or dividers and use the horizontal scale to find out the separation in ppm. The conversion is then easy—to turn parts per million of megahertz into hertz you just leave out the million! So 1 ppm on a 300 MHz machine is 300 Hz. On a 500 MHz machine, a 10 Hz coupling is a splitting of 0.02 ppm.

- Spectra from different machines

When you change from one machine to another, say, from a 200 MHz to a 500 MHz NMR machine, chemical shifts (δ) stay the same in ppm and coupling constants (J) stay the same in Hz.



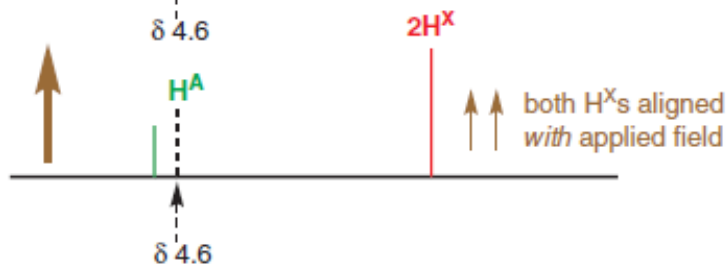
If one proton H^A interacts with two protons H^X , it can experience protons H^X in three different possible states. Both protons H^X can be aligned with the magnet or both against. These states will increase or decrease the applied field just as before. But if one proton H^X is aligned with and one against the applied field, there is no net change to the field experienced by H^A . There are two arrangements for this. We'll therefore see a signal of double intensity for H^A at the correct chemical shift, one signal at higher field and one at lower field. In other words, a 1:2:1 triplet.

effect of H^A coupling to two protons H^X

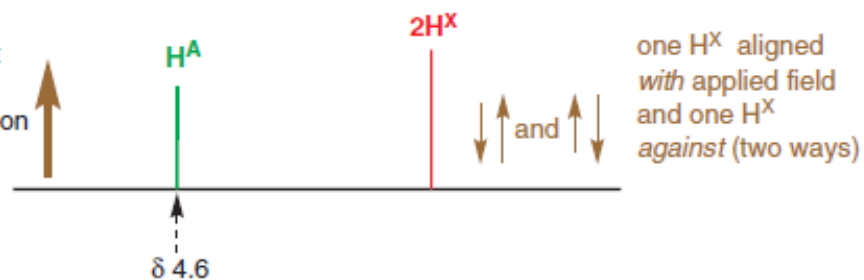
spectrum with
no interaction



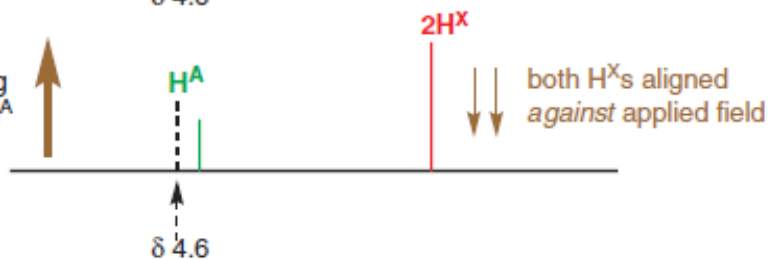
effect of H^X and
applied field acting
together on H^A



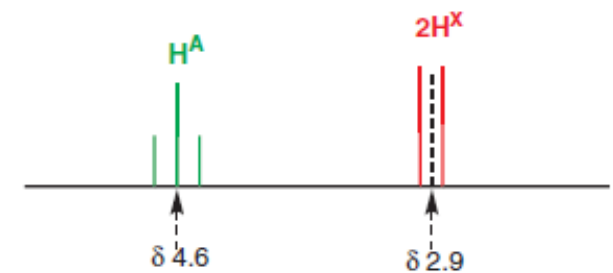
two H^X s cancel out
double intensity
signal at true position



effect of H^X and
applied field acting
in opposition on H^A



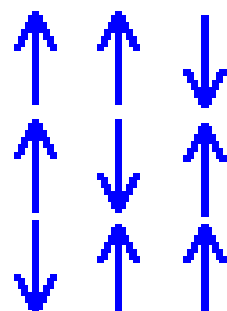
resultant
spectrum:
effect of H^X on H^A
and of H^A on H^X



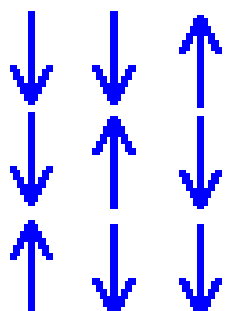
spin on 3H



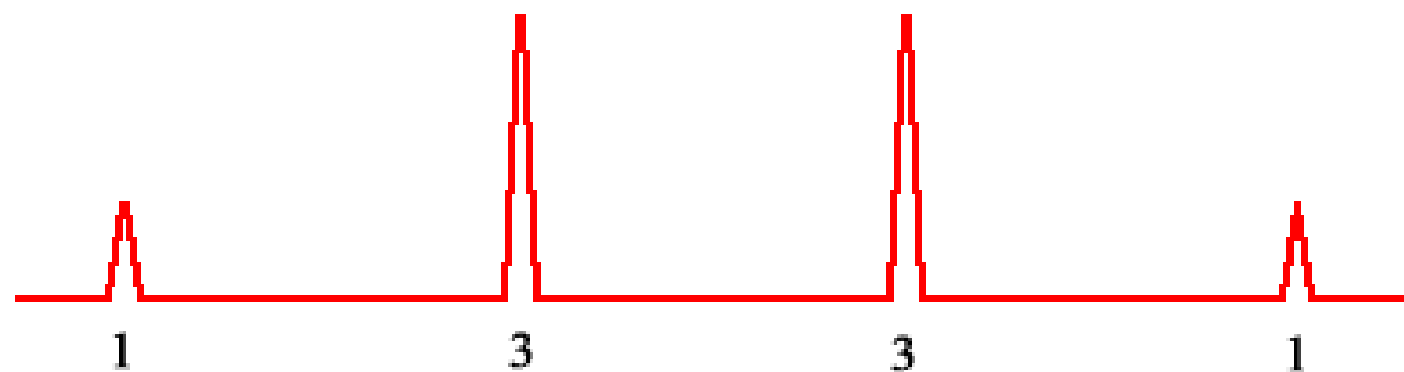
spin on 3H



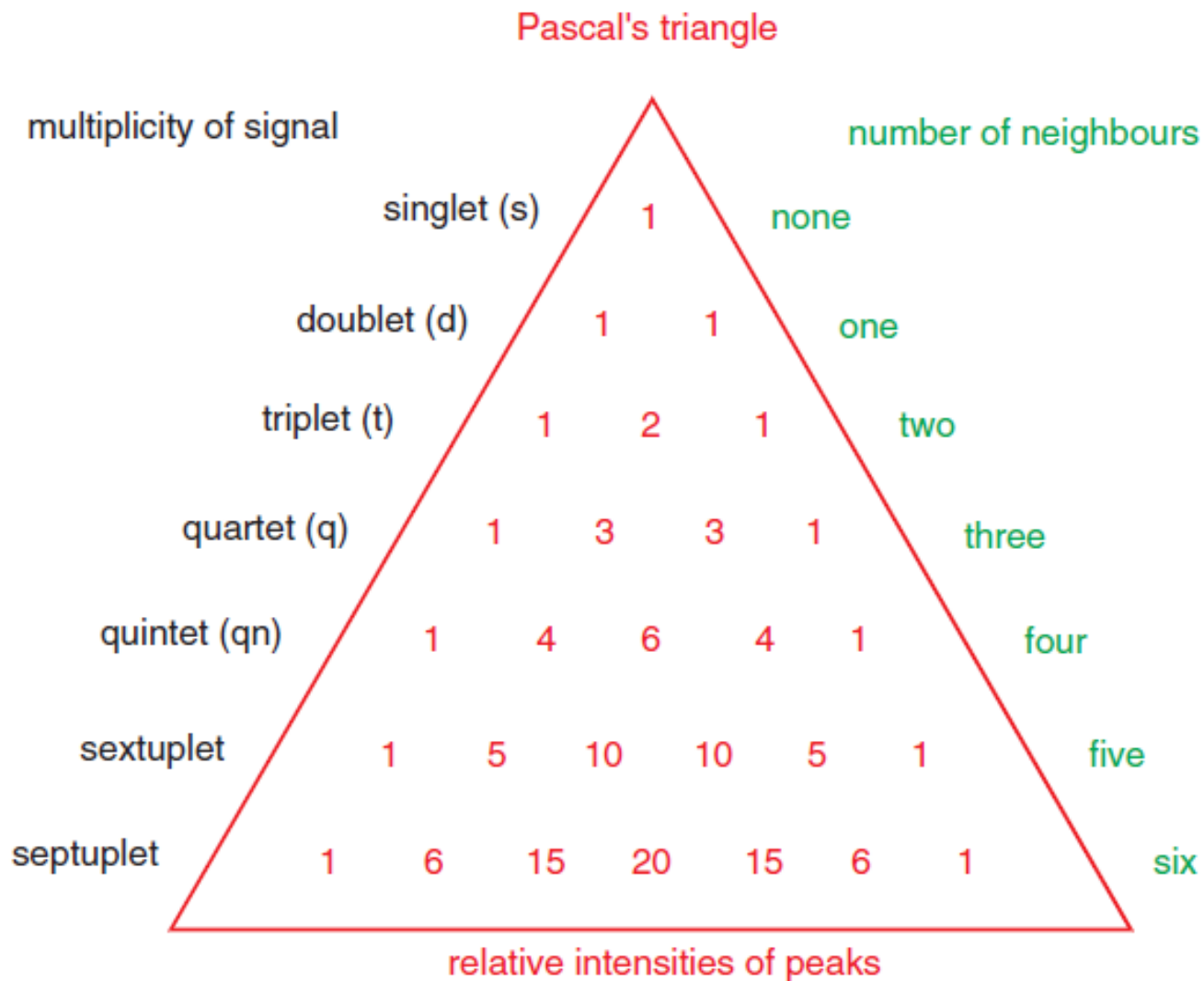
spin on 3H



spin on 3H

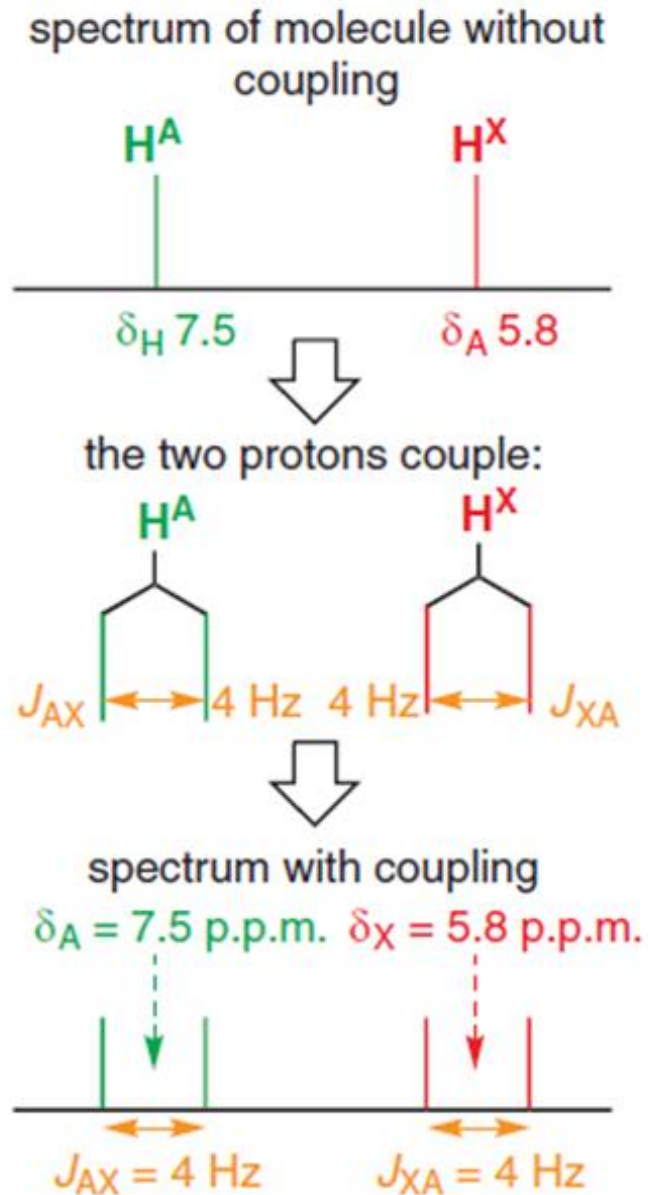


quartet

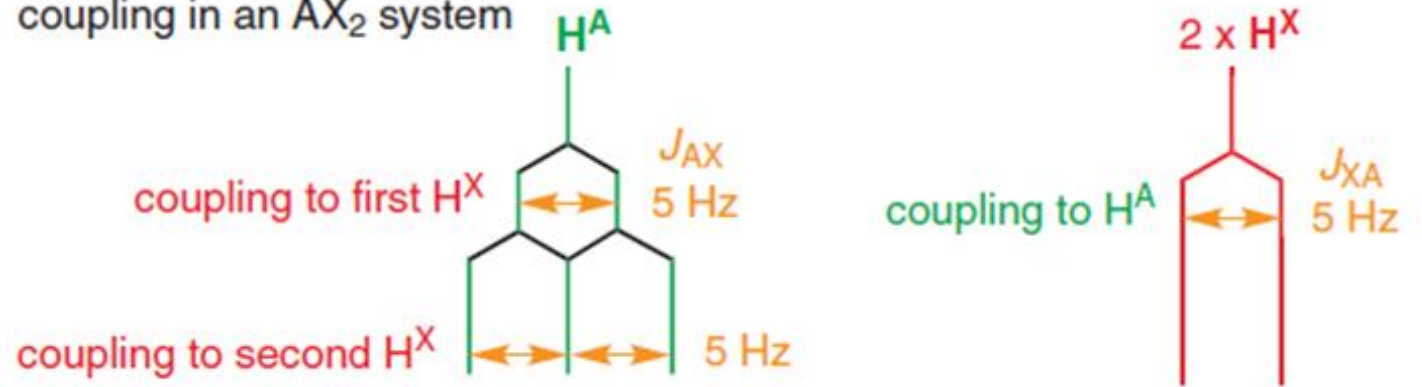


Constructing Pascal's triangle: Put '1' at the top and then add an extra number in each line by adding together the numbers on either side of the new number in the line above. If there is no number on one side, that counts as a zero, so the lines always begin and end with '1'.

Third explanation:



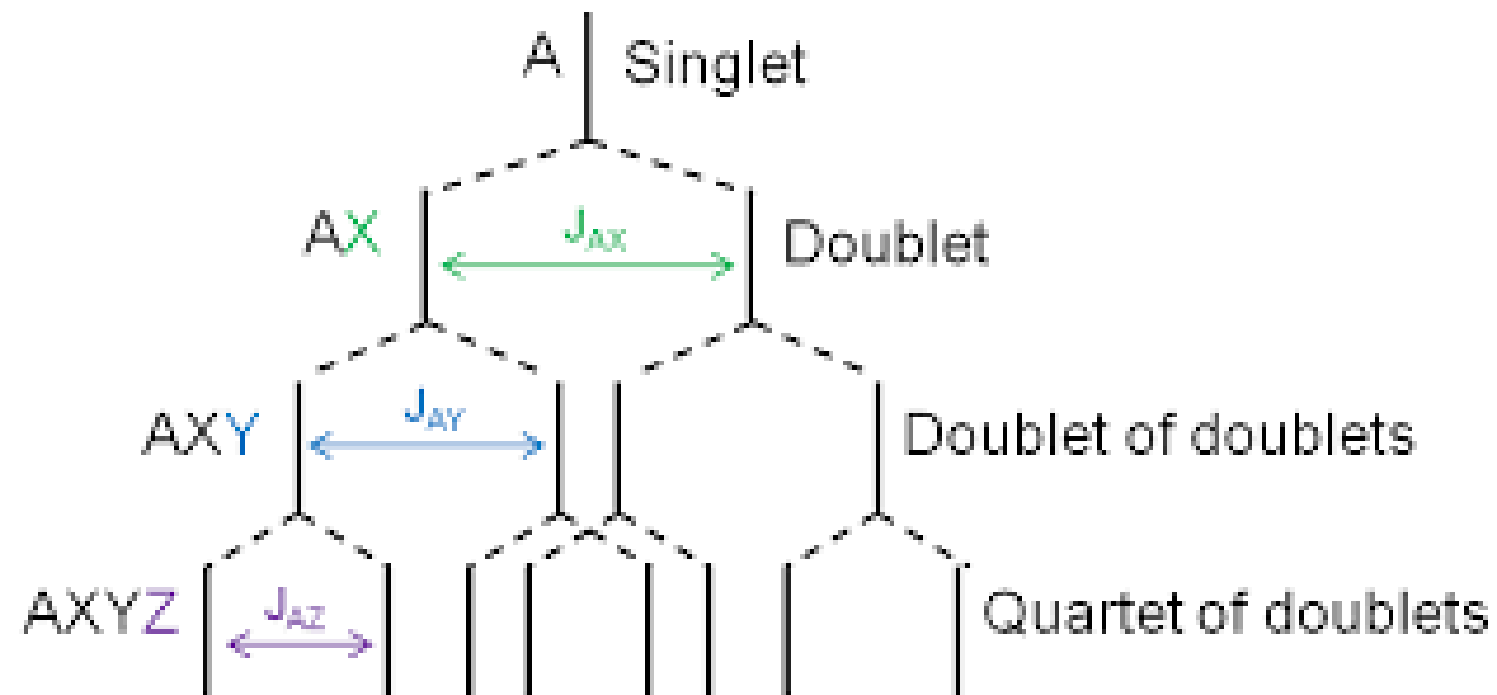
coupling in an AX_2 system

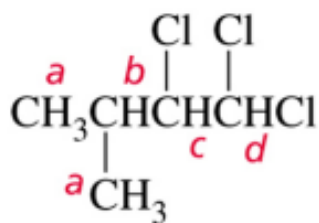


the resulting AX_2 spectrum

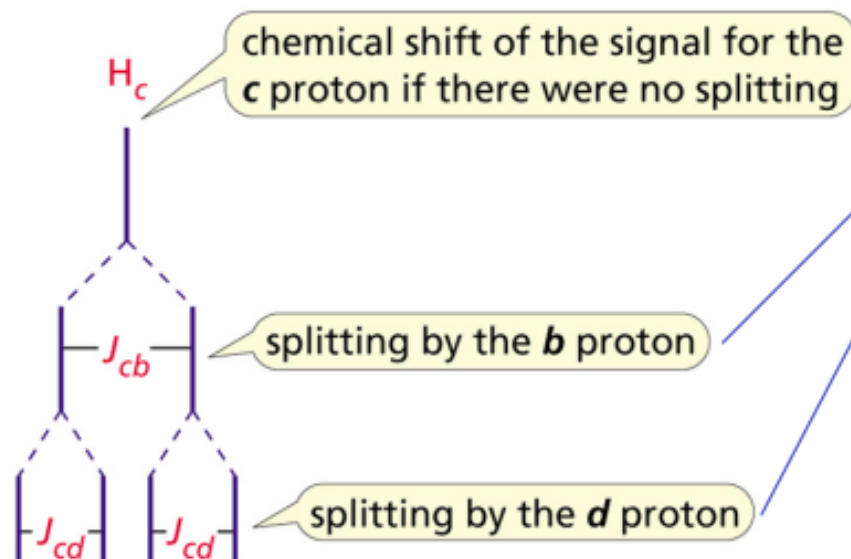
(brown numbers show relative peak intensities)







a splitting diagram

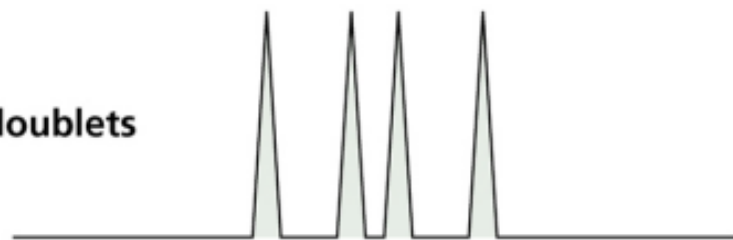


c split by **b** and **d**
b and **d** not equiv
 $\rightarrow 2 \times 2$

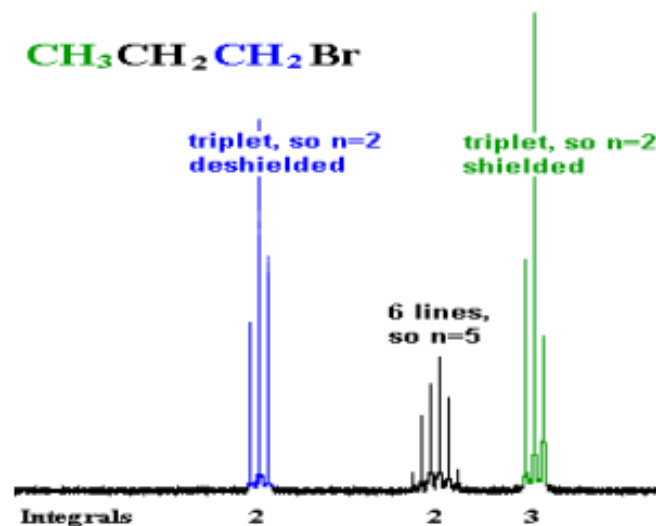
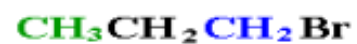
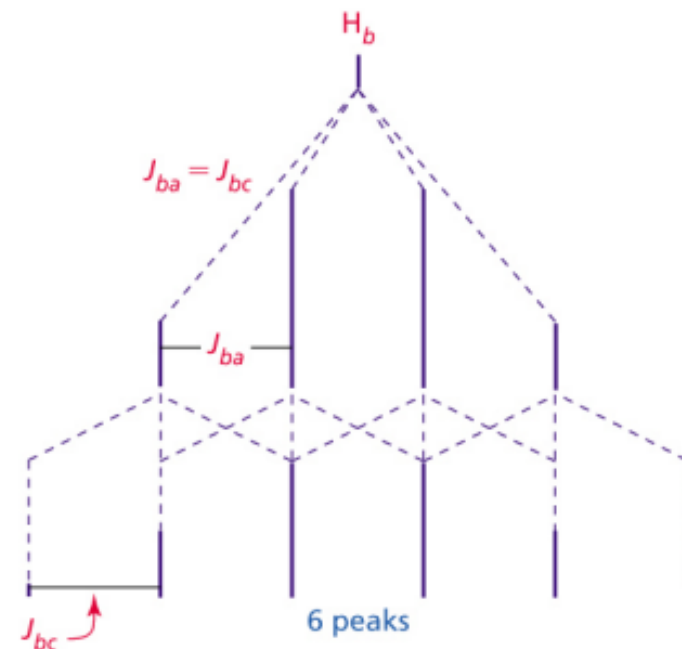
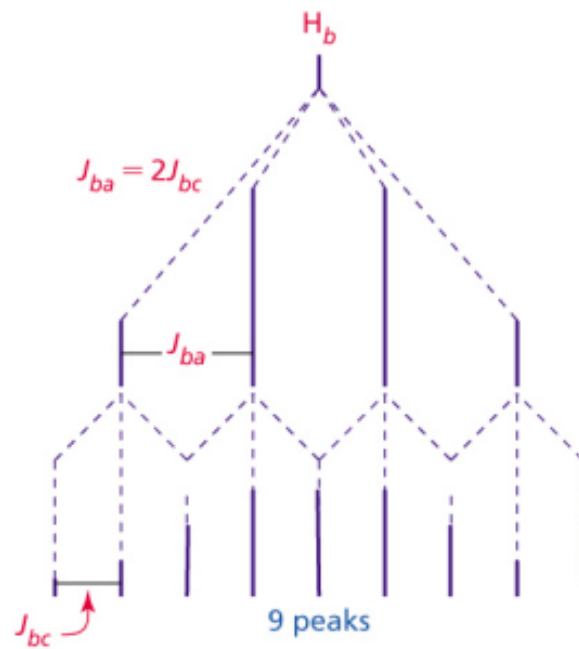
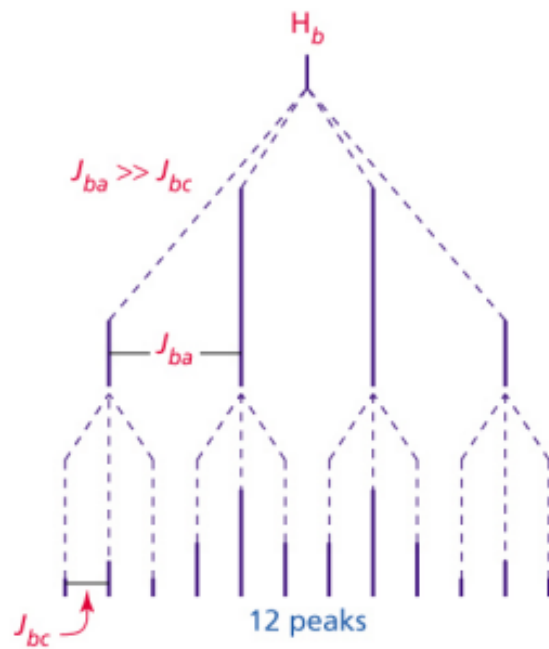
split by **b**
 then by **d**

$J_{cb} \neq J_{cd}$
 (**b**, **d** not equiv)
 $\rightarrow$ all seen

a doublet of doublets



if $J_{cb} \approx J_{cd}$
 (**b**, **d** equiv or somehow)
 $\rightarrow$ not all seen



**a and c not equiv
but J's close enough
~ results the same**